

Daily Content Intel

July 27, 2026

Topic A — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, so the old concussion advice was rest in a dark room until you feel normal again. The research moved.

There's a randomized trial in PLOS One, adolescents and young adults after a sport related concussion, and one group got a structured aerobic protocol starting inside the first week. 8 sessions over 11 days, 20 minutes each, riding at 60 to 75% of their age predicted max heart rate.

The exercise group hit asymptomatic status faster, and they got medically cleared earlier, and their symptom scores were lower at every single check-in.

Here's the mechanism, probably. A concussion messes with how the brain regulates blood flow and autonomic function, and light aerobic work kept below the symptom threshold seems to nudge that system back toward normal instead of letting it sit.

Now I'll hedge, because a 2026 trial in Physical Therapy ran this in adults who were already 3 to 24 months out from their injury, and symptom scores didn't budge, though exercise tolerance did improve. So timing matters here. Early is where the win is.

What that means Monday. If one of my athletes gets a concussion, we test their symptom threshold, and we get them on a bike that same week, supervised, staying under that threshold.

Does that make sense?

Be Good. Help Someone. Learn Lots.

Hook: Sitting in a dark room after a concussion is the slow way back.

Spoken script (word for word):

Okay, so the old concussion advice was rest in a dark room until you feel normal again. The research moved.

There's a randomized trial in PLOS One, adolescents and young adults after a sport related concussion, and one group got a structured aerobic protocol starting inside the first week. 8 sessions over 11 days, 20 minutes each, riding at 60 to 75% of their age predicted max heart rate.

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On-screen text seeds:

- 0:00 The dark room is the slow way back
- 0:08 PLOS One RCT: aerobic exercise vs usual care after concussion
- 0:16 8 sessions / 11 days / 20 min
- 0:22 60% to 75% of age-predicted max HR
- 0:30 Faster to asymptomatic. Earlier medical clearance.

- 0:40 Below the symptom threshold, not through it
- 0:52 2026 caveat: starting late didn't move symptoms
- 1:02 Early is where the win is
- 1:10 Test the threshold. Get on the bike that week.

Caption (copy-paste ready):

The dark room advice is out of date.

A randomized trial in PLOS One put adolescents and young adults on a structured aerobic protocol inside the first week after a sport related concussion. 8 sessions over 11 days, 20 minutes each, at 60 to 75% of age predicted max heart rate. That group reached asymptomatic status faster, got cleared earlier, and reported lower symptom scores at every follow up.

The honest caveat: a 2026 trial in Physical Therapy tried this in adults who were 3 to 24 months out and symptom burden didn't change, though exercise tolerance did. Early is where the win is.

If your athlete gets a concussion, the question is what their symptom threshold is, and how soon you can start working underneath it.

Be Good. Help Someone. Learn Lots.

Dr. Lars Stevenson, PT, DPT

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#concussion #returntoplay #footballtraining
#lacrosse #highschoolfootball
#sportsmedicine #athletictraining
#physicaltherapy #youthsports
#returntosport

Personalize this

- Have you managed a concussion return-to-play with one of your HS football players, and what did the first week actually look like compared to what the school or the ATC wanted?
- What's your own way of finding an athlete's symptom threshold when you don't have a Buffalo treadmill protocol available in the clinic?
- Where does this land in CROSS? Talk about how you'd sequence the bike work

alongside the neck and vestibular pieces before contact clearance.

Screenshots:

- title | <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0276336> | "Randomized controlled trial of early aerobic exercise following sport-related concussion: Progressive percentage of age-predicted maximal heart rate versus usual care"
- physiology | <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0276336> | "Compared to the UCEP group, the SAEP had a faster time to asymptomatic status with 96% posterior probability."
- actionable | <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0276336> | "An aerobic exercise protocol based on percentages of age-predicted maximum heart rate is a safe and effective treatment for reducing symptoms and can be initiated during the first week following SRC."

Sources:

- <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0276336>
- <https://academic.oup.com/ptj/article/106/6/pzag049/8675871>

Topic B — Reel

TELEPROMPTER COPY (paste into Instagram)

Okay, hot take for the coaches. Chronological age is a garbage way to group a youth team, and the growth spurt is where you're losing kids to injury.

There's an integrative review out this year in Research, Society and Development, 31 studies, U-11 all the way through U-23, and the conclusion is blunt. Maturational stage should replace approaches based solely on chronological age.

A Delphi consensus of academy practitioners landed on the same window. The period during peak height velocity and the 12 months after it, along with strength and flexibility imbalances, are the most important maturity related injury risk factors they deal with.

Here's what's actually happening in that kid. The long bones lengthen ahead of the soft tissue, so the levers get longer while the muscle and the tendon are still catching up, and coordination goes sideways for a while. That's the phase everybody calls lazy.

So I'd track height every 6 to 8 weeks. When a kid jumps more than 3 centimeters in a couple of months, back the volume off, keep the strength work in, and stop asking for a personal record every single week.

The growth spurt is a training variable. Program it.

Be Good. Help Someone. Learn Lots.

Hook: Stop cutting the kid who got clumsy over the summer.

Spoken script (word for word):

Okay, hot take for the coaches.

Chronological age is a garbage way to group

a youth team, and the growth spurt is where you're losing kids to injury.

There's an integrative review out this year in Research, Society and Development, 31 studies, U-11 all the way through U-23, and the conclusion is blunt. Maturational stage should replace approaches based solely on chronological age.

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Here's what's actually happening in that kid. The long bones lengthen ahead of the soft tissue, so the levers get longer while the muscle and the tendon are still catching up, and coordination goes sideways for a while. That's the phase everybody calls lazy.

So I'd track height every 6 to 8 weeks. When a kid jumps more than 3 centimeters in a couple of months, back the volume off, keep the strength work in, and stop asking for a

personal record every single week.

The growth spurt is a training variable.
Program it.

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On-screen text seeds:

- 0:00 Stop cutting the clumsy kid
- 0:07 Chronological age is a bad grouping tool
- 0:15 31 studies, U-11 to U-23
- 0:22 "Maturation stage should replace chronological age"
- 0:32 Riskiest window: during PHV + 12 months after
- 0:42 Bones lengthen first. Soft tissue catches up late.
- 0:52 Track height every 6 to 8 weeks
- 1:00 >3 cm in 2 months = pull volume, keep strength
- 1:08 The growth spurt is a training variable

Caption (copy-paste ready):

The 8th grader who suddenly can't cut, can't land, and looks a step slow is not a motivation problem.

An integrative review this year pulled 31 studies across U-11 to U-23 and concluded that maturational stage should replace approaches based solely on chronological age. A Delphi consensus of academy practitioners put the highest maturity related injury risk in the window during peak height velocity and the 12 months after it, alongside strength and flexibility imbalances.

The long bones lengthen before the soft tissue catches up. Longer levers, same coordination software, and it takes a while to recalibrate.

Track height every 6 to 8 weeks. A jump over 3 centimeters in a couple of months means you pull volume, keep the strength work, and stop chasing a PR every week.

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#youthsports #biobanding
#peakheightvelocity
#strengthandconditioning #footballtraining
#lacrosse #highschoolathletes
#injuryprevention #athleticdevelopment
#longtermathleticdevelopment

Personalize this

- In the HS football program you write, how do you actually adjust for the sophomore who grew 4 inches between spring ball and camp, and what does his week look like versus the senior next to him?
- Have you had a lacrosse athlete land in your clinic with Osgood-Schlatter or a growth-related complaint that the coach had been reading as effort, and how did you explain it to the parents?
- What's your honest position on bio-banding for the NoVA club scene, and what would you tell a club director who says it hurts college exposure?

Screenshots:

- title |
<https://rsdjournal.org/rsd/article/view/50542>

| "Biological maturation, training load, and injury risk in youth soccer: An integrative review on performance, exercise-induced adaptations, and safe sport development strategies in young athletes"

- physiology | <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0312568> | "Panellists agreed that the period during and 12-months post peak height velocity, muscle strength/flexibility imbalances and maturity status (% predicted adult height) as the most important maturity-related injury risk factors."
- actionable | <https://rsdjournal.org/rsd/article/view/50542> | "It is concluded that considering maturational stage is essential for safe and effective training prescription and should replace approaches based solely on chronological age in youth soccer contexts."

Sources:

- <https://rsdjournal.org/rsd/article/view/50542>

- <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0312568>

Reference

Runner-up topics

- FIFA 11+ and ankle injury reduction in football players (systematic review + meta-analysis, PMC12371935). A warm-up that actually moves ankle injury rates, and most teams cut it short.
- Prior hamstring strain leaves persistent morphology and sprint mechanics deficits in NCAA D1 football players (Scand J Med Sci Sports 2026, paywalled, need an open mirror). Strong argument that "pain-free" is a bad clearance criterion.
- In-season minimum effective dose: one hard weekly session held strength in adolescent male athletes as well as two. Good hot take against the 4-day in-season lift.

Sources scanned today

- **Newsletter digest**
(newsletter_digest_2026-07-27.pdf, 3 sources): Dr. Mercola x2 (supplement routine promo, gut microbe promo) and Mike T. Nelson ("Greetings from OH", a

personal Table Talk podcast story). No usable clinical finding. Skipped as promo/personal.

- **PubMed / journal sweep** (athlete-biased terms: high school, collegiate, football, lacrosse, in-season, sprint, return to sport): surfaced in-season minimum-dose lifting, sprint volume and hamstring eccentric strength, cold water immersion and hypertrophy, creatine safety in adolescents, sub-symptom aerobic exercise after concussion, maturation and injury risk.
- **Dedup rejections (worth noting):** creatine safety in adolescents (Rubinchuk 2026 Cureus) and cold water immersion blunting hypertrophy were both fully drafted, then dropped. Creatine already ran as Draft A on 2026-06-17 and Draft B on 2026-07-02; cold plunge ran Draft B on 2026-05-27 and Draft A on 2026-06-29. Fresh URLs, stale topics.
- **Also passed over:** in-season minimum effective dose (adjacent to strength-training-beats-conditioning-for-speed, 2026-07-24), sprint volume and hamstring

strength (covered 2026-07-17 and 2026-07-22), FIFA 11+ ankle injury meta-analysis (held as runner-up).

- **Unreachable / paywalled:** Wiley (Scand J Med Sci Sports hamstring morphology study, HTTP 402), Wiley (EJSS cold water meta-analysis, HTTP 402), Cureus HTML article body (JS-gated; read via the open-access PDF and the PMC mirror instead), Ovid PDF for S&C Journal 48(3).